

RAYALA AKHIL

Backend Engineer | Python • AWS • Kubernetes • DevOps Automation

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PROFILE SUMMARY

Backend Engineer with 3+ years of experience building large-scale security automation systems at Cisco (via HCL). Specialized in Python-based ETL pipelines, scan orchestration across 5,000+ assets, and cloud-native deployments on AWS + Kubernetes. Seeking high-impact product engineering roles.

PROFESSIONAL EXPERIENCE

Software Engineer – Cloud Security Engineering Operations (Cisco)

Jun 2024 – Present

HCL Technologies Ltd., Bengaluru, India

- Orchestrated automated security scans across **10,000+** Cisco assets, reducing manual scan effort by **~70%** through Python-based scheduling and workflow automation.
- Built ETL/data processing pipelines for scan results, enabling structured reporting and actionable insights for security teams – cutting report generation time from hours to minutes.
- Implemented real-time alerting system (Webex + Email) notifying 10+ security team members, reducing incident response lag by **~40%**.
- Designed **10+** CRON-based schedulers for periodic workflows, reducing manual intervention from **~20** hrs/week to near-zero.
- Integrated AWS (S3, DynamoDB) for storage and metadata management, Redis for caching and fast lookups, and SharePoint for report distribution.
- Containerized and deployed microservices on Kubernetes; managed pod lifecycle, deployments, and operational monitoring – maintaining **99%+** uptime.
- Authored CI/CD pipelines via GitLab and automated routine operations using Bash and Python scripts across Linux (Ubuntu) environments.

Technical Lead – Internationalization (i18n)

Aug 2022 – May 2024

HCL Technologies Ltd., Chennai, India

- Led i18n efforts for Cisco products vManage and NDO, ensuring compliance across multiple locales.
- Built internal automation tools to streamline i18n validation checks and testing workflows, reducing QA turnaround by **~30%**; recognised with internal reward for value-added contributions.

PERSONAL PROJECTS

Web-Based Excel Automation Tool

- Built a full-stack Flask web application to help my father's business workflow that automates multi-sheet Excel parsing and reporting; integrated Supabase (PostgreSQL & Storage) to persist file history and ConvertAPI to support automatic PDF generation.
- Reduced a 3–4 hour manual Excel calculation process to under 10 seconds (**~98%** time reduction) by building the web application using NumPy and OpenPyXL
- Containerized the application with Docker and set up a GitHub Actions CI/CD pipeline to auto-build and push Docker images to Docker Hub on every commit or pull request.

TECHNICAL SKILLS

Languages: Python, Bash, Java (basic), JavaScript

Cloud & Infrastructure: AWS (S3, DynamoDB), Kubernetes, Docker, Redis

Automation & ETL: CRON, Workflow Orchestration, Scan Orchestration, Data Pipelines, Scripting

DevOps & CI/CD: GitLab CI/CD, GitHub Actions, Docker Hub, Linux (Ubuntu)

APIs & Integrations: Webex API, Email APIs, SharePoint, REST APIs

Frontend : React, Angular

Data & ML (coursework): NumPy, Pandas, Matplotlib, Scikit-learn

Domains: SecOps Automation, Microservices, Backend Engineering, Cloud Infrastructure

AI & Generative Tools:

- AI-Assisted Development: Extensively utilized Cursor and GitHub Copilot to accelerate software development workflows, including boilerplate generation, debugging, and unit test validation.
- AI Application Development: Built and deployed a functional chatbot application using OpenAI APIs and React, managing backend API integration and frontend state management.
- Prompt Engineering: Refined prompt strategies to improve AI model output accuracy for code refactoring and troubleshooting complex logic.
- Lifecycle Integration: Applied AI-driven coding assistance throughout the software development lifecycle to accelerate documentation, test case generation, and architectural prototyping.

CERTIFICATIONS

CCNA – Cisco Certified Network Associate

EDUCATION

Bachelor of Technology – Materials Science and Engineering

Indian Institute of Technology, Kanpur | Kanpur, India

May 2018 – Apr 2022

- Secretary, Astronomy Club | Member, Programming Club

COURSES & CONTINUOUS LEARNING

- Machine Learning – Coursera: Fundamentals of supervised and unsupervised learning.
- Data Structures and Algorithms in Python – Udemy
- Data Analysis with Python – freeCodeCamp: Worked on large Kaggle datasets using NumPy, Pandas, Seaborn, and Matplotlib.